INCIDENT REPORT

Enterprise SOC Detection Lab — Ransomware Simulation & Splunk Cloud Detection Engineering

Incident Ref	IR-SPLUNK-2026-001	**Classification**	LAB SIMULATION
Date	October 5, 2026	**Severity Rating**	**CRITICAL (Level 12)**
Investigating Analyst	Idris Lawan Mustapha	**SIEM Platform**	Splunk Cloud Platform
Target Endpoint	DESKTOP-F95RRHE (192.168.238.129)	**Attacker Node**	parrot-htb (192.168.238.128)

1. Executive Summary

This formal incident report documents an end-to-end simulated cyberattack and corresponding detection engineering response executed against an enterprise Windows 10 endpoint within an isolated VMware environment. The simulation replicated the multi-stage tactics, techniques, and procedures (TTPs) characteristic of modern ransomware operators (e.g., LockBit, BlackCat, Akira), culminating in pre-encryption recovery inhibition and credential theft.

Telemetry was continuously collected at the kernel level via Microsoft Sysmon v15.21 and transmitted over mTLS to the **Splunk Cloud Platform** (prd-p-o1pgk.splunkcloud.com) using the Splunk Universal Forwarder. An initial baseline and gap analysis demonstrated that standard out-of-the-box SIEM searches failed to escalate destructive administrative recovery inhibition commands (such as vssadmin delete shadows) into actionable high-severity alerts. In response, five custom high-fidelity SPL detections were engineered, scheduled, and validated to ensure immediate alerting and MITRE ATT&CK; contextualization.

2. Lab Architecture & Network Topology

Node Role	Hostname / OS	IP / MAC Address	Security / Attack Stack
Threat Actor	parrot-htb (Parrot OS Linux)	192.168.238.128	Metasploit v6.4.136, Nmap 7.95, Python HTTP
Victim Target	DESKTOP-F95RRHE (Win 10 x64)	192.168.238.129 00:0C:29:B8:EF:28	Sysmon v15.21, Splunk UF v10.4.3
SIEM Cloud	Splunk Cloud Instance	prd-p-o1pgk.splunkcloud.com	Search & Reporting, Alert Manager, Dashboards

3. Incident Attack Timeline & Verified Evidence

Timestamp	Attack Phase	Observed Command / Telemetry	MITRE
2026-10-04 14:55 UTC	Baseline Telemetry	Sysmon service active (Event ID 16, 4 verified)	Baseline
2026-10-04 15:27 UTC	Pipeline Active	Splunk UF connected to prd-p-o1pgk.splunkcloud.com:9997	Ingest
2026-10-04 16:50 UTC	Payload Execution	Dropped binary executed: C:\Users\Public\payload.exe	T1204.002
2026-10-04 17:02 WAT	Network Recon	Full port sweep via Nmap: sudo nmap -sV -O -p- 192.168.238.129	T1046
2026-10-04 17:03 UTC	Host Recon	powershell -ExecutionPolicy Bypass -Command 'Get-LocalUser'	T1059.001
2026-10-04 17:04 UTC	Defense Evasion	powershell -Command 'Set-MpPreference -DisableRealtimeMonitoring $true'	T1562.001
2026-10-04 17:50 WAT	C2 Socket Opened	Meterpreter reverse TCP established (victim :51329 -> attacker :4444)	T1571
2026-10-04 18:12 WAT	Privilege Escalation	getsystem via named pipe impersonation -> NT AUTHORITY\SYSTEM	T1134
2026-10-04 18:12 WAT	Credential Theft	hashdump extracted local SAM hashes (Admin, FIKIRA, Guest)	T1003
2026-10-05 16:00 UTC	Recovery Inhibition	vssadmin.exe delete shadows /all /quiet executed	T1490
2026-10-05 16:38 UTC	Alert Engineering	5 custom scheduled SPL alerts enabled in Splunk Cloud	Defense
2026-10-05 17:15 UTC	Alerts Triggered	CRITICAL - Ransomware Staging alert fired in Splunk Activity Console	T1490

4. MITRE ATT&CK; Mapping & Engineered Detections

ID	Tactic	Technique	Engineered Detection Rule	Severity
T1046	Reconnaissance	Network Service Scanning	DET-001 (Sysmon EID 3 port fan-out count)	Medium
T1204.002	Execution	Malicious File	DET-002 (Public/Temp path execution)	High
T1059.001	Execution	PowerShell Abuse	DET-003 (Bypass / Hidden / Disable flags)	High
T1003.001	Credential Access	LSASS Memory Snooping	DET-004 (Sysmon EID 10 handle inspection)	High
T1562.001	Defense Evasion	Disable Security Tools	DET-005 (Set-MpPreference tampering)	High
T1490	Impact	Inhibit System Recovery	DET-006 (vssadmin / bcdedit / wbadmin)	CRITICAL

5. Indicators of Compromise (IOCs)

Indicator Type	Observed Artifact / Value	Context / Notes
Attacker IP	192.168.238.128	Source of Nmap scan and Metasploit C2 listener
Victim IP / MAC	192.168.238.129 / 00:0C:29:B8:EF:28	Target enterprise endpoint (DESKTOP-F95RRHE)
C2 Socket	192.168.238.128:4444	Outbound TCP connection from victim port 51329
Malicious File	C:\Users\Public\payload.exe	Unsigned Meterpreter reverse shell binary (7,680 bytes)
Recovery Inhibition	vssadmin delete shadows /all /quiet	Executed via elevated CMD to purge shadow copies
Ransom Artifacts	README_DECRYPT.txt FINANCIAL_2026.xlsx.enc	Simulated extortion note and encrypted file dropped on desktop

6. Containment & Strategic Recommendations

1. Immediate Host Isolation: Disconnect the compromised endpoint (DESKTOP-F95RRHE) from the network to terminate active reverse TCP sockets and prevent lateral movement.
2. Restrict World-Writable Directories: Deploy AppLocker or Software Restriction Policies to prevent user-space binaries from executing from C:\Users\Public and user AppData\Local\Temp paths.
3. Lock Down Administrative Utilities: Restrict execution of vssadmin.exe, bcdedit.exe, and wbadmin.exe strictly to approved administrative groups.
4. Enforce Tamper Protection: Enable Tamper Protection via Microsoft Intune or Group Policy to neutralize scripts attempting to disable antivirus monitoring via Set-MpPreference.
5. Production Deployment of Custom SPL Detections: Ingest all engineered alert queries (DET-001 through DET-006) into Splunk Cloud production pipelines with automated webhook notifications to SOC messaging channels.

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